



Review article

Prevention of cardiovascular disease for healthy aging and longevity: A new scoring system and related “mechanisms-hallmarks-biomarkers”

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ABSTRACT

Healthy “environment-sleep-emotion-exercise-diet-intellectualization [E(e)SEEDi] lifestyle can improve the quality of life, prolong aging and promote longevity due to improvement of human immunity and prevention of cardiovascular diseases (CVD). Here, the author reviewed the associations between these core elements with CVD and cardiovascular aging, and developed a new scoring system based on the healthy E(e)SEEDi lifestyle for prediction and evaluation of life expectancy. These core factors are assigned 20 points each (120 points in total), and a higher score predicts healthier aging and longevity. The E(e)SEEDi represents “a tree of life” bearing the fruits of longevity as well as “a rocket of anti-aging” carrying people around the world on a journey of longevity. In conclusion, the E(e)SEEDi can delay aging and increase the life expectancy due to the role of a series of cellular and molecular “mechanisms-hallmarks-biomarkers”. It is believed that the novel scoring system has a huge potential and beautiful

1. Introduction

Life expectancy in the United States is expected to decline in the 21st century because of the impact of obesity on longevity (O'Carroll et al., 2005). Khan et al (Khan et al., 2018). reported that patients with obesity were associated with shorter lifespan and significantly higher morbidity and mortality rates compared to those with normal body mass index (BMI). This suggested that a healthy lifestyle was required to prevent obesity and other risk factors. Nilsson et al (Carlsson et al., 2020). reported that the life expectancy of Swedish obese subjects was significantly increased by bariatric surgery to control BMI. In clinical trials, treatment with therapeutic drugs such as liraglutide (a glucagon-like peptide-1 analogue) and chloroquine (a selective serotonin-2C receptor agonist) was associated with significant weight loss in patients with obesity, especially when combined with lifestyle changes (Pi-Sunyer et al., 2020; Shuhua et al., 2018). However, the COVID-19 infection had the potential to affect the global economic and social development, but also global life expectancy (Islam et al., 2021; Pifarré I Arolas et al., 2021).

According to the 2019 Global Health Estimates of the World Health

Organization (WHO), (The World Health Organization, 2020) seven major non-communicable diseases (mNCDs) (ischaemic heart disease; stroke; chronic obstructive pulmonary disease; neonatal conditions; trachea, bronchus and lung cancers; Alzheimer's disease and other forms of dementia; diabetes) were among the top 10 causes of death worldwide compared to 4 mNCDs (ischaemic heart disease; stroke; chronic obstructive pulmonary disease; bronchus and lung cancers) in the year 2000. Therefore, in order to achieve the United Nations Sustainable Development Goals and increase human life expectancy, novel strategies are required to prevent or treat mNCDs, including cardiovascular diseases (CVD), diabetes, cancer, respiratory and kidney diseases, and obesity whose incidence rates are constantly increasing in most countries (GBD, 2015). Japan has the highest life expectancy in the world because of the robust primary and secondary preventive measures against mNCDs and efficient community public health measures (Ikeda et al., 2011).

The recently released 2020 report on Nutrition and Chronic Diseases in China showed that the prevalence of hypertension, diabetes, cancer, and obesity was 27.5 %, 11.9 %, 0.2939 %, and > 50 %, respectively. The average life expectancy in China was 77.4 years in 2019 and 78.2 in

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Table 1 (continued)

Core Elements	Contents Associated with Healthy Aging & Longevity	Score Value
	food (coffee, milk, tea with antioxidant, trace elements Se, Zn); None or low alcohol drinking, no smoking; Prescription drugs (statin, traditional Chinese medicine).	

Notes: The total score is 120. The higher the score, the longer the human life expectancy. ApoE: apolipoprotein E; BMI: body mass index; BPIFB4: bactericidal/permeability-increasing fold-containing family B member 4; C282Y: Cys282Tyr; CETP: Cholesteryl ester transfer protein; d3-GHR: growth hormone receptor polymorphism with deletion of exon 3; DASH: Dietary Approaches to Stop Hypertension; E(e)SEEDi: Environment-Sleep-Emotion-Exercise-Diet intervention; eNOS: endothelial nitric oxide synthase; EQ: emotional intelligence; HLA: human leukocyte antigen; IGF-1: insulin-like growth factor-1; IL-6: interleukin 6; IQ: intelligence quotient; KLOTHO: rs1207568 and rs564481; PGC1 α : peroxisome proliferator-activated receptor gamma coactivator-1-alpha; OSA: obstructive sleep apnea.



Fig. 1. A life tree with fruits of longevity”: The healthy E(e)SEEDi lifestyle-based scoring system. E(e)SEEDi refers to the “environment (external & internal)-sleep-emotion-exercise-diet intervention” and is represented as “a life tree with fruits of longevity”. E(e)SEEDi was first described by Dr. Chunsong Hu, and includes six healthy lifestyle-based core elements that delay aging and promote longevity. E(e)SEEDi prolongs cardiovascular aging and promotes longevity, especially in patients with chronic heart failure. A novel scoring system was developed for the prediction and evaluation of life expectancy and longevity. E (left): External environment; e: Internal environment; S: Sleep; E (upper): Emotion; E (right): Exercise; D: Diet; i: Intervention.

et al., 2019) showed better performance for aging and longevity, but did not predict the initial telomere length alone. Furthermore, aging is related with the composition of the gut microbiota (Bjork et al., 2019). The regulation of gut microbiota enhances immunity and longevity because nearly 70 % of immune cells are located in and around the intestinal tract (Hu, 2020).

Secondly, high-quality sleep and daily half-hour naps increase immunity, reduce mNCDs, and prevent acute events in humans. Thirdly, optimism and good personality traits are modifiable biomedical factors that promote longevity (Lee et al., 2019; Stieger et al., 2021). Self-control in childhood and adulthood, (Richmond-Rakerd et al., 2021) and a good mental attitude including low stress levels and worries contribute to longevity. Since psychological stress is associated with CVD, relaxation techniques (meditation and listening to music) can improve microvascular function, decrease inflammation, and increase longevity (Dal Lin et al., 2018, 2021). In fact, studies have found that happiness increased the longevity and physiological health of

individuals (Frey, 2011).

Fourthly, exercise and physical activity are crucial factors that contribute to longevity. Individuals engaged in receptive arts activities such as going to museums, operas, art galleries, etc, were associated with increased longevity because of improved cognitive and mental health (Fancourt and Steptoe, 2019). Furthermore, aerobic activity and lifelong endurance exercise were associated with increased longevity (Koch et al., 2011; Aengevaeren et al., 2017; Parry-Williams and Sharma, 2020). A calorie- and glucose-restricted diet was associated with healthy aging and longevity (Fontana et al., 2010; Fontana and Partridge, 2015; Heintz et al., 2017; Zou et al., 2020) because it improved mood, QoL and sleep in the adults (Martin et al., 2016). Furthermore, a diet rich in high-quality protein, vitamins, and trace elements increased longevity (Ames, 2011; Hebert et al., 2015) by improving immunity (Annas and Galis, 2018). The intake of plant protein extends life expectancy by reducing the risk of death from all causes and CVD (Naghshi et al., 2020). Therefore, a healthy diet is closely related with health and longevity.

Natural products and traditional Chinese medicine (TCM) are an integral part of daily life. For example, the extract of *B.monniieri* prolonged the health span of *C.elegans* by decreasing mitochondrial and oxidative stress (Grinson et al., 2020). Several natural products and TCM regimens (Lin et al., 2011) such as green tea, (Yang et al., 2014) sesame, (Lin et al., 2014) mulberry fruit, (Huang et al., 2013; Chaiwong et al., 2021) and some cereja species (summer savory) (Hamidpour et al., 2021) have shown beneficial biological properties including antioxidant, anti-inflammatory, and antibiotic properties against microbial infections, hypoglycemic, hypolipidemic, anti-atherosclerosis, and others. Therefore, natural products and TCM promote healthy aging and longevity by preventing or effectively treating mNCDs such as CVD, type 2 diabetes (T2D), cancer, and Alzheimer’s disease.

Prevention of CVD for human healthy aging and longevity

The world population is aging rapidly and there is significant inequality in life expectancy based on social and economic factors worldwide (Mackenbach et al., 2019). Therefore, the WHO has released a policy framework to guide a coherent and focused public health response to promote healthy aging (Beard et al., 2016). CVD is one of the leading causes of mortality worldwide and is a roadblock against healthy aging and longevity. The number of CVD cases nearly doubled from 271 million in 1990–523 million in 2019 (Roth et al., 2020). Therefore, positive control and prevention of CVD is a vital step for healthy aging and longevity. Furthermore, several cardiovascular health biomarkers (Table 1), such as BMI, (Khan et al., 2018) systolic BP, (Simoes et al., 2021) and absolute lipoprotein (a) levels (Arsenault et al., 2020) are also related to longevity in humans. Therefore, the aim of the healthy E(e)SEEDi lifestyle and drugs is to normalize these CVD-related indicators in order to achieve healthy aging and longevity.

Healthy aging and longevity are the main goals of the modern civilization. The lifespan setpoint for a particular species determines the life expectancy of any given organism (Lidsky and Andino, 2020). However, healthy lifestyle may drive evolution of the lifespan setpoint. A rational design of strategies for improving longevity in the complex aging system requires regular application of therapeutics and engineered system maintenance (Sun et al., 2020). Castillo-Quan et al (Castillo-Quan et al., 2019). demonstrated that a combination of established life-extending drugs (trametinib, rapamycin and lithium) increased the lifespan of *Drosophila* by 48 %. The author postulates that the healthy E(e)SEEDi lifestyle-based scoring system can predict and assess life expectancy and longevity.

4. Related “mechanisms-hallmarks-biomarkers” and future prospects

Research on centenarians in China confirmed the role of a healthy E

(e)SEEDi lifestyle in promoting longevity by suppressing mNCDs and improving the QoL and human immunity. Recent clinical studies have confirmed that increased intake of fresh fruits and vegetables (Wang et al., 2021a) and diets with a low glycemic index (Jenkins et al., 2021) are beneficial for cardiovascular health and longevity. The positive and healthy E(e)SEEDi lifestyle can play a key role in human life expectancy by protecting telomere length (Pizzo, 2020; Arbeev et al., 2020; Crous-Bou et al., 2014).

Several studies have demonstrated the protective effects of sirtuins (SIRT) such as SIRT1, SIRT3, SIRT6, and SIRT7 in CVD (atherosclerosis, dyslipidemia, cardiomyopathies, and myocardial infarction) and diabetes (Cardus et al., 2013; Winnik et al., 2015; Sosnowska et al., 2017; Onofrio N et al., 2018). These longevity genes protect against vascular aging-related pathology and prolong human life expectancy and health. A better understanding of the CVD-related major cellular and molecular “mechanisms-hallmarks-biomarkers” of human aging and longevity, and the developments of new drug targets (Cannatà et al., 2016) and novel interventional strategies are necessary for increasing longevity. The healthy E(e)SEEDi lifestyle components including exercise and caloric restriction protect against early cardiac and vascular aging by activating the AMP-activated protein kinase (AMPK) and increasing the endothelial expression and function of SIRT1 (Wang et al., 2018; Guo et al., 2019).

The IDF Diabetes Atlas 2021 (Tenth Edition) (International Diabetes Federation, 2021) reported that more than 3 out of 4 adults among the 537 million adults living with diabetes worldwide reside in low- and middle-income countries. In 2021, diabetes accounted for 6.7 million deaths and cost nearly \$966 billion US dollars in medical expenditure (International Diabetes Federation, 2021). The delivery of care for the aging population with diverse physiological changes and complications involves multi-morbidity and chronic disease management and the application of evidence-based medicine to improve the longevity and clinical outcomes of the elderly subjects, especially those with CVD (Dowling et al., 2020; Power et al., 2013; Jennings et al., 2006). Genetic variants that alter the metabolic pathways influence life expectancy by mediating interactions with several nutritional and environmental factors, but their effects are significantly reduced at an advanced age. However, a healthy lifestyle and other environmental factors can significantly modify the outcomes of aging (Finch and Hall, 2013). For example, during the COVID-19 pandemic, the use of surgical masks significantly influenced the exercise capacity of adult male and male participants compared to that of females (Alkanani et al., 2020).

Healthy aging is key for enhancing longevity (López-Otín and Rowe, 2020; Dzau et al., 2019). Advances (Table 2) in the understanding of “mechanisms-hallmarks-biomarkers” related to cardiovascular development and injury and human aging (Liu et al., 2020; Zhang et al., 2015) and the discovery of novel therapeutic targets and methods, (Zhang et al., 2020; Wang et al., 2021b; Min et al., 2021) have shown evidence that life expectancy can be prolonged by delaying cardiovascular aging. For example, several studies have clearly demonstrated the association between cardiovascular aging and derailment of proteostasis, which includes homeostasis of protein synthesis, folding, assembly, trafficking, function, and degradation (Wiersma et al., 2016). Therefore, novel and more effective treatment strategies are now available for preventing or delaying the progression of CVD (eg, atrial fibrillation) and improving survival outcomes. Further understanding of the epidemiological features of CVD (Zhao, 2021) and the pathophysiological mechanisms associated with aging and longevity (Pietri and Stefanadis, 2021) will unravel better clinical strategies against cardiovascular aging and promote longevity and healthy aging in “cells-tissues-organs (CTO)” (Fig. 2 & Fig. 3). All in all, this new scoring system will meet the increasing demand of high quality of cost-effective health care services for healthy aging and longevity in the new era. The increasing knowledge about healthy aging and longevity can narrow the educational and occupational gap worldwide and promote healthy aging in individuals belonging to different socioeconomic groups.

Table 2
Current Spectrum of Major Cellular and Molecular “Mechanisms-Hallmarks-Biomarkers” of Human Aging and Longevity.

No	Major Cellular and Molecular Mechanisms/Hallmarks of Aging and Longevity	Major Cellular and Molecular Biomarkers of Aging and Longevity
1	Altered intercellular communication	GDF-15, IL-6, CRP, TNF-α
2	Apoptosis, ferroptosis, necrosis	dsDNA, ncRNAs, caspase activity, IL-6, TNF-α, caspase3/9, cleaved caspase-3
3	Cellular senescence	GDF-15, dsDNA, CD4 + /CD8 + Rate, p16, p21/p16, p53/p21, SASP, Lamin B1 & LRP1
4	Deregulated nutrient-sensing	IGF-1
5	Disabled macroautophagy	LC3, Circular RNAs, ncRNAs, microRNAs, nucleic acids (nucleotides)
6	Dysbiosis (Microbiome disturbance)	Mucosa, OPA1, Drp1
7	Epigenetic alterations	Global DNA methylation
8	Genomic instability	Mitochondrial DNA copy number
9	Inflammation	IL-1β, IL-6, TNF-α, iNOS
10	Loss of proteostasis	Hsp70 and Hsp72
11	Metabolic unbalance	Autonomic nervous system (ANS)
12	Mitochondrial dysfunction	Mitochondrial DNA copy number, IL-15
13	Oxidative stress	IL-1β, IL-6, CRP, TNF-α, NF-κB, dsDNA
14	Stem cell exhaustion	Circulating progenitor cells
15	Cellular attrition	Telomere length
16	Unfolded protein response (UPR) (endoplasmic reticulum [ER] stress)	miR-34 family (miR-34a, miR-34b, or miR-34c) of microRNAs/miRNAs (including E2F1, c-Myc, and cyclin E2, miR-122-5p, miR-17-5p and miR-125b-5p), transmembrane protein 135 (Tmem135), fibroblast growth factor 21 (FGF21), growth differentiation factor 15 (GDF15), humanin (HN), and transcription factor 4, BiP, XBP1

Notes: Because these mechanisms/hallmarks are linked, interventions of genetic pathways and biochemical processes that ameliorate one hallmark/biomarker can in theory ameliorate others. Major noncommunicable diseases (mNCDs) include cancer, diabetes, cardiovascular disorders, neurodegenerative diseases, aging, obesity, metabolic syndrome, chronic inflammation, and other diseases. Targeting cells include cardiomyocytes, endothelial cells, vascular smooth muscle cells, fibroblasts, immune cells, progenitor cells, and others. These candidate hallmarks and their relative contributions to aging are worthy of research, and the final goal is identifying the healthy E(e)SEEDi lifestyle or pharmaceutical targets to improve human health during aging, with no/minimal side effects. These evidences are from both review-based and research-based literatures.

In fact, there have been a series of review-based evidences on major cellular and molecular “mechanisms-hallmarks-biomarkers” (Pereira et al., 2023; Tchkonina et al., 2021; López-Otín et al., 2013, 2023; Abdellatif et al., 2023) of human ageing and longevity (Table 2), and therapeutic strategies (Raffin et al., 2023; Rebelo-Marques et al., 2018; Shannon et al., 2021; Grande de França et al., 2023; Chaib et al., 2022) of anti-aging based on the healthy E(e)SEEDi lifestyle for CVD, (Suda et al., 2023; Sweeney et al., 2023) as well as research-based evidences (Tyshkovskiy et al., 2023; Debès et al., 2023; Soerens et al., 2023) on mechanisms of longevity and interventions by targeting RNA polymerase II and functional T cells. With more studies on a pluripotent or senescent cells, (Liu and Qu, 2022; Sun et al., 2022) flavonoids-natural compounds, (Taylor et al., 2022; Fan et al., 2022) nicotinamide mononucleotide (NMN) (Soma and Lalam, 2022), and taurine (Singh et al., 2023; McGaunn and Baur, 2023; Vidal Valero, 2023) as well as mammals (companion dogs, mice or mouse population), (Cagan et al., 2022; Creevy et al., 2022; Acosta-Rodríguez et al., 2022; Bou Sleiman et al., 2022) people will further disclose the major cellular and molecular “mechanisms-hallmarks-biomarkers” of human ageing and

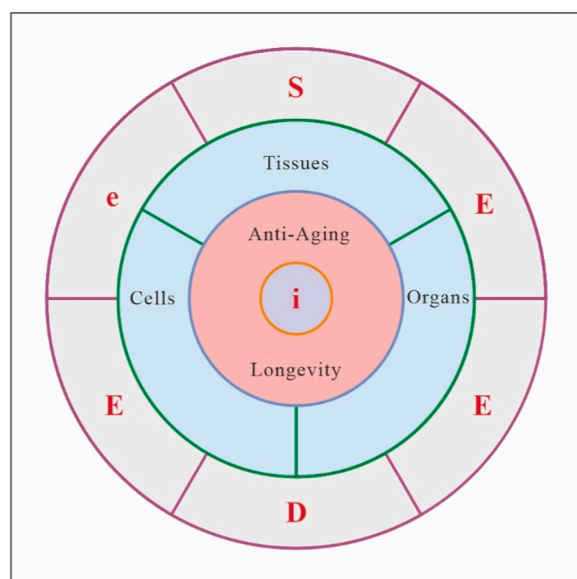


Fig. 2. The healthy E(e)SEEDi for anti-aging and longevity in cells, tissues, and organs (CTO). Here, E (left): External environment; e: Internal environment; S: Sleep; E (upper): Emotion; E (right): Exercise; D: Diet; i: Intervention.

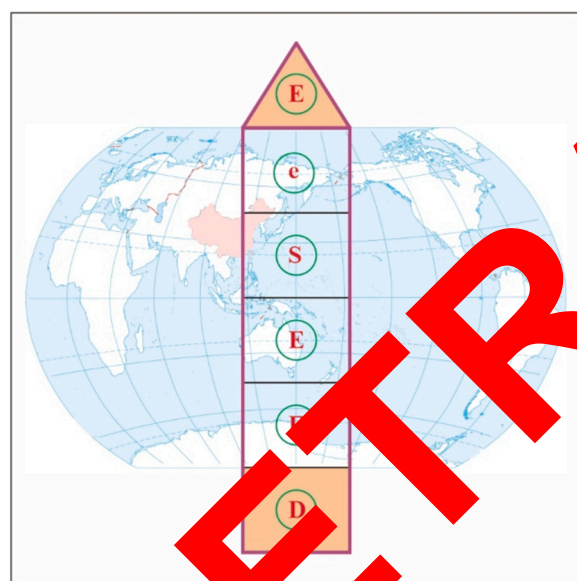


Fig. 3. The healthy E(e)SEEDi is just like a basket, carrying people around the world on a journey of anti-aging and longevity. Here, E (upper): External environment; e: Internal environment; S: Sleep; E (middle): Emotion; E (down): Exercise; D: Diet; i: Intervention.

longevity with many novel technologies, (Longo and Anderson, 2022; Green et al., 2022; Gami et al., 2022; Allen et al., 2023) and better prevention of cardiovascular aging and infection-related CVD by “Hot Pots” of both traditional Chinese medicine and marine natural products for longevity in the pandemic and post-COVID-19 era (Hu, 2023a, 2023b, 2024).

Since some factors in the blood (such as factor V, VII, X, and ABO blood group) link to longevity, (Faure-Delanef et al., 1997; Meiklejohn et al., 2000; Girolami et al., 2005; Franchini et al., 2016) control of infection and inflammation (COVID-19), (Schulman et al., 2020) assaying the biomarker (P-selectin) level, (Jing et al., 2018) ethnic genetic analysis (Tang and Hu, 2015) and evaluation of hereditary and lifestyle (acquired) risk factors, (Zeng et al., 2015) will help to prevent

venous thromboembolism (VTE), thrombotic risk in AF, and ischaemic stroke risk, (Guo et al., 2016; Chao et al., 2022) innovative strategies (for example, machine learning and artificial intelligence) (Guo, 2022) are helpful to prevent these cardiovascular risk factors and get more chances of longevity. At the same time, literature from *Ageing Research and Reviews* (Mattson et al., 2017; Brown-Borg and Buffenstein, 2017; Chiang et al., 2020; Denham and Sellami, 2021; Olivieri et al., 2023; Galkin et al., 2023; Miceli et al., 2023; Morgan and Mc Auley, 2024) also confirm and support the role of healthy lifestyle on human health, ageing and longevity.

5. Conclusion

The healthy E(e)SEEDi lifestyle is a simple option for improving the QoL and human immunity of individuals belonging to different age groups, including those of advanced age. “The Essential 5” of this healthy lifestyle can prevent or suppress diseases including CVD, T2D, and cancer, and show a significantly high correlation with cardiovascular health, cardiac aging, and longevity. This novel scoring system can be used worldwide to prevent and assess cardiovascular aging, healthy lifespan, and human longevity. In conclusion, the healthy E(e)SEEDi lifestyle can obviously delay cardiovascular aging and increases the human life expectancy. And the author believes that this novel scoring system has a large potential and beautiful prospects.

Author contributions

CH conceived, planned, drafted, edited, and revised the manuscript. The author has read and approved the final version of the manuscript for submission.

Declaration of Competing Interest

The author declare that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.arr.2025.102727](https://doi.org/10.1016/j.arr.2025.102727).

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